Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Development of a High-Multiplex Biomarker Imaging Platform for Precision Diagnostics and Therapeutic Innovation			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1 (**Large Category**)**		**Technology Field 2 (**Large Category**)**	
	Bio / Medical Technology		AI / Digital Healthcare / Intelligent Medical Devices	
	Medium Category	Small Category	Medium Category	Small Category
	Biomedical Imaging & Diagnostics	Molecular Imaging / Multiplex Biomarker Imaging	Digital Pathology	AI-enabled Biomarker Analysis
Concept of Proposed Technology	The proposed technology is an advanced imaging platform based on the principle of Fluorescence Lifetime Multiplexing (FLEX), which combines spectral fluorescence information with fluorescence lifetime measurements to enable quantitative and high-level multiplex biomarker visualization. This approach overcomes the inherent limitations of conventional immunofluorescence imaging, which is typically constrained by spectral overlap and limited multiplexing capacity. The FLEX platform supports simultaneous imaging of up to twelve biomarkers and also provides label-free metabolic imaging capability. These features allow comprehensive analysis of the structural and molecular characteristics of tissues, enabling detailed assessment of complex tumor microenvironments.			

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Rationale for Joint R&D Project	The system integrates optical design, imaging hardware, and analytical software to support a broad range of applications in biomedical research, drug discovery, and precision diagnostics. As a reliable and scalable imaging platform, FLEX has the potential to contribute significantly to the advancement of precision and personalized medicine. This project aims to advance the development and translation of the FLEX imaging platform by integrating academic, clinical, and industrial expertise. Building on extensive collaborative research and validated proof-of-concept studies, the initiative seeks to accelerate the transition of FLEX technology from research innovation to practical application in precision and personalized medicine. The ultimate goal is to establish FLEX as a robust, scalable, and accessible solution for both clinical and research settings. The consortium brings together complementary capabilities spanning biological and clinical validation, system engineering, scalable fabrication, and data analytics. Clinical and academic partners contribute expertise in translational research, regulatory guidance, and clinical evaluation, ensuring that the technology meets stringent standards for safety, performance, and usability. Technical and industrial collaborators provide system optimization, optical and engineering integration, and pathways for technology transfer and scale-up. This multidisciplinary approach strengthens the project's capacity to accelerate technology readiness and adoption. A key objective of the project is to enable iterative improvement of FLEX technology through real-world testing, end-user feedback, and rigorous performance validation. Collaborative studies with healthcare institutions will facilitate refinement of imaging protocols, assessment of clinical workflows, and generation of evidence supporting clinical and translational value. The consortium also places emphasis on addressing regulatory and quality compliance requirements to pave the way for eventual certification and implementation in diverse medical environments. Successful completion of this project will position FLEX technology to contribute meaningfully to the advancement of precision diagnostics and tailored therapeutics. By fostering international and cross-sector collaboration, the initiative demonstrates a model for how coordinated R&D partnerships can accelerate medical technology innovation and enhance healthcare outcomes worldwide.
Key Words	1. Highly Multiplexed Biomarker Imaging 2. Spatial Biology 3. Fluorescence Lifetime Imaging 4. Precision Medicine 5. Companion Diagnostics

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (4)	End of the Project	Level (6)
R&D Objectives	• Optimize FLEX imaging instrumentation to improve reliability, scalability, and suitability for routine research and clinical applications. • Enhance multiplex biomarker detection capabilities, including sensitivity, specificity, and throughput, to enable comprehensive tumor biology investigations. • Integrate advanced software analytics and AI-assisted data interpretation to facilitate biological discovery and clinical decision-support. • Collaborate with clinical teams at Massachusetts General Hospital to ensure translational relevance through identification of clinically meaningful biomarker panels, validation with annotated tumor specimens, and evaluation of workflow compatibility within pathology and research settings. • Generate supporting evidence for potential companion diagnostic applications, emphasizing patient stratification, therapy response prediction, and tumor microenvironment analysis from a clinical perspective.			
R&D Target Specifications	• Capability for single-shot, 12-plex simultaneous biomarker imaging. • Quantitative lifetime-based signal differentiation with minimal spectral and temporal overlap. • High-resolution tissue imaging fully compatible with standard pathology workflows. • Consistent and reproducible performance demonstrated across multi-site validation studies.			
R&D Plan	**Mission 1: System Robustness & Engineering Optimization** • Improve optical stability, fluorescence lifetime signal sensitivity, and multiplex accuracy to ensure consistent performance across diverse tissue types. • Enhance detector electronics, timing resolution, calibration routines, and system automation to support routine, reliable operation. • Optimize mechanical design, thermal stability, and system alignment to ensure long-term reliability and robustness. • Expand software automation, including image acquisition workflows, multiplex unmixing algorithms, and quantitative analysis pipelines. • Establish standardized operating procedures, calibration protocols, and quality assurance frameworks to support reproducible system operation. **Mission 2: Expanded Translational Validation in Clinical Research Settings** • Conduct systematic translational validation studies using clinically annotated tumor specimens. • Benchmark platform performance against established pathology techniques, including IHC, IF, and spatial biology platforms. • Validate high-plex biomarker imaging capability (approximately 12-plex) and label-free metabolic imaging under real-world research conditions. • Evaluate workflow integration within pathology laboratories, including sample preparation, throughput, usability, and data interpretation. • Generate peer-reviewed publications demonstrating analytical validity, clinical relevance, and translational utility. **Mission 3: Multi-Site Deployment & Operational Demonstration** • Deploy systems across multiple academic and pharmaceutical research environments to enable broad-based validation.			

	• Assess reproducibility, robustness, and performance consistency across institutions. • Collect real-world user feedback to refine usability, system reliability, and software interfaces. • Demonstrate operational readiness in translational research environments that reflect future clinical adoption. • Strengthen datasets supporting companion diagnostic development and precision oncology applications. **Mission 4: Pre-Commercial Engineering & Manufacturing Readiness** • Advance pre-commercial engineering efforts focused on manufacturability optimization, modular system design, and cost reduction strategies. • Establish supply chain pathways, hardware reliability testing protocols, and quality control frameworks. • Expand the software ecosystem to include AI-assisted analysis, data integration, and interoperability with digital pathology systems. • Prepare technical and operational documentation supporting regulatory strategy and commercialization pathways. **Mission 5: Commercial Translation & Partnership Expansion** • Expand strategic partnerships with pharmaceutical and biotechnology stakeholders for biomarker discovery and companion diagnostic research. • Develop service-based offerings to enable early revenue generation while preparing for full instrument commercialization. • Build global collaboration networks spanning academia, clinical research centers, and industry. • Prepare investor-facing validation packages, clinical evidence summaries, and commercialization strategy documentation.

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	**Academic Clinical Partner (Massachusetts General Hospital / Harvard Medical School Collaboration)** • Massachusetts General Hospital (MGH) is a globally recognized academic research hospital and a major hub for translational biomedical innovation, providing a strong clinical validation environment for FLEX technology. • Provide access to clinically relevant tumor tissue samples, translational research cohorts, and clinical insight to guide biomarker panel development and validation studies. • Lead clinical study design, biological interpretation, and evaluation of FLEX performance in real-world oncology research environments. • Benchmark FLEX against existing pathology and multiplex imaging technologies to establish clinical utility and differentiation. • Support publication of peer-reviewed studies at major scientific conferences, and validation in clinically meaningful contexts. • Identification of clinically meaningful biomarker panels and use cases that inform future companion diagnostic development. • Provide feedback on workflow integration, usability, pathology compatibility, and future regulatory considerations to facilitate clinical translation.
Expected R&R of Korean SME Partner	• Lead development of core FLEX technologies, including fluorescence lifetime optical system architecture, multiplexed imaging instrumentation, and end-to-end hardware–software platform integration and optimization. • Leverage industry-standard expertise in automated visual inspection, precision manufacturing, and optical engineering to accelerate hardware development, optics integration, automation, and scalable production readiness. • Advance engineering refinement to ensure system reliability, robustness, and manufacturability, encompassing detector optimization, optical alignment, system automation, validation, and quality assurance frameworks. • Develop proprietary data analysis pipelines, AI-assisted image processing algorithms, and user-facing software interfaces tailored for research, translational, and future clinical workflows. • Drive manufacturing scale-up, supply chain development, cost optimization, and global commercialization readiness through established industrial manufacturing and operations practices. • Establish and execute intellectual property strategy, including patent development, branding, and protection of trade secrets related to system integration, algorithms, workflows, and platform differentiation. • Coordinate regulatory documentation preparation, manufacturing transfer, and commercialization strategy execution to support translational and clinical deployment pathways. • Lead market development activities, including customer discovery, strategic partnerships with pharmaceutical and biotechnology stakeholders, pilot system deployments, and preparation for companion diagnostic and precision medicine applications.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	This collaboration is positioned to deliver a validated, high-multiplex tumor imaging platform that will significantly enhance biomarker discovery, accelerate therapeutic development, and advance precision oncology. Through close engagement with leading clinical research environments, the project ensures strong clinical relevance and direct applicability of the technology to patient-derived samples, thereby improving translational fidelity and real-world impact. Partnership with industrial and manufacturing stakeholders supports efficient technology translation and commercialization, enabling scalable production and readiness for global deployment. Key expected outcomes include the generation of high-impact scientific publications, the strengthening of international research collaborations, and the creation of new pathways for global R&D exchange. The project is designed to unlock commercial opportunities such as diagnostic kit development and advanced data analytics services for pharmaceutical, biotechnology, and healthcare stakeholders, supporting sustainable growth and long-term adoption. The FLEX platform is expected to emerge as a benchmark technology in cancer diagnostics and research, with broad implications for clinical decision-making and therapeutic innovation. Ultimately, these efforts aim to improve cancer patient outcomes worldwide by enabling more accurate diagnosis and personalized treatment strategies.